Semester Project Report

THRUST-BASED BEAM BALANCING USING BARE-METAL CONTROL

MCT-336L – Embedded Systems-II Lab
(Fall-2025)

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Submitted by: **Group #: 13B**

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"ABSTRACT"

➢ This semester project presents the design and implementation of a thrust-based beam balancing system using bare-metal control on an ARM Cortex-M4 microcontroller. The system consists of a single-axis beam pivoted at its center and stabilized using differential thrust generated by two brushless DC (BLDC) motors mounted at the beam ends. A Texas Instruments TM4C123G (Tiva C Series) microcontroller operating at 16 MHz is used to execute a real-time feedback control loop. Beam pitch angle is measured using an MPU6050 inertial measurement unit (IMU) interfaced via I2C, while motor speed and current are monitored using Hall-effect sensors and ACS712 current sensors respectively. Pulse-width modulation (PWM) signals are generated to control electronic speed controllers (ESCs) driving the motors. Serial communication is implemented to transmit system parameters to a PC for monitoring and future plotting. The project demonstrates the application of embedded systems, sensor interfacing, and feedback control concepts for stabilizing an inherently unstable mechanical system.

"INTRODUCTION"

- **Brief Overview:**

➢ Balancing and stabilization problems are fundamental in control engineering and embedded systems due to their presence in numerous real-world applications such as drones, robotic platforms, camera gimbals, and self-balancing vehicles. These systems are inherently unstable and require continuous real-time feedback to maintain equilibrium. The integration of sensors, actuators, and control algorithms under strict timing constraints makes such problems both challenging and educationally valuable.

This project focuses on the implementation of a one-dimensional beam balancing system using thrust as the actuation mechanism. Unlike traditional servo-based balancing systems, thrust-based stabilization introduces additional complexities such as nonlinear motor dynamics, actuator delays, and safety considerations. The system is controlled using a bare-metal firmware approach on the TM4C123G microcontroller, allowing direct interaction with hardware peripherals without relying on high-level libraries or an operating system. This approach provides deeper insight into low-level embedded system design and real-time control implementation.

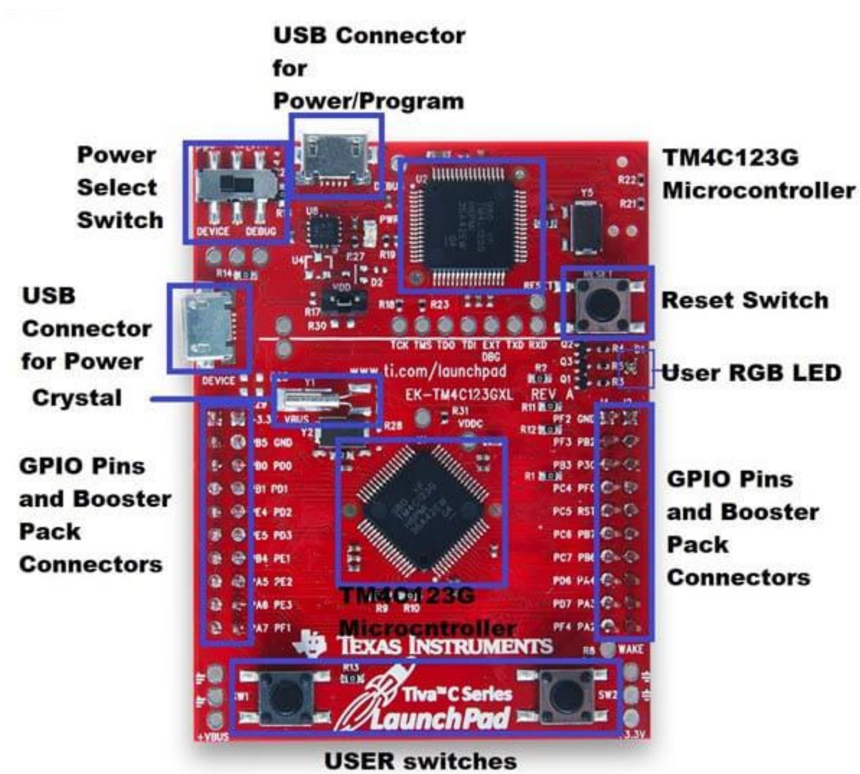
- **<u>Applications</u>:**

  - **Unmanned Aerial Vehicles (UAVs) and Drones:**
    Differential thrust from multiple motors is used to control pitch, roll, and yaw angles. The same PID-based feedback control and IMU sensing used in this project are fundamental to drone flight stabilization.
  - **Camera Gimbals and Stabilization Systems:**
    Single-axis and multi-axis gimbals use sensor feedback and motor actuation to counter vibrations and unwanted motion, ensuring smooth and stable video capture.
  - **Self-Balancing Robots and Vehicles:**
    Systems such as two-wheeled robots and personal transporters rely on real-time angle measurement and corrective actuation to maintain upright balance, similar to the beam balancing problem.
  - **Industrial Automation and Active Leveling Systems:**
    Precision machines and robotic platforms use feedback control to maintain alignment and compensate for disturbances, improving accuracy and operational safety.
  - **Aerospace Attitude Control Systems:**
    Satellites, missiles, and launch vehicles use thrust or torque modulation with sensor feedback to maintain desired orientation during operation.

“COMPONENT ANALYSIS”

1. Microcontroller (Tiva C Series TM4C123GH6PM):

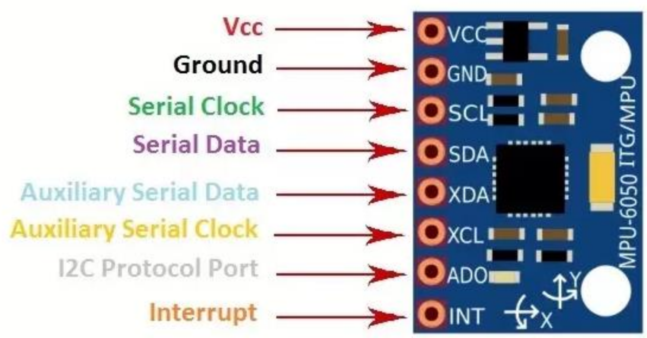
- **Description:** A 32-bit ARM Cortex-M4 based microcontroller with a wide range of digital and analog peripherals for embedded applications.
- **Role:** Serves as the central processing unit of the system. It reads sensor data from the IMU, Hall-effect RPM sensors, and current sensors, executes the control algorithm, and generates PWM signals to control the BLDC motors through ESCs.
- **Interaction:** Interfaces with the MPU6050 via I2C for angle measurement, controls ESCs using PWM outputs on GPIO pins, receives RPM feedback via Hall sensors, reads current values through ADC channels, and communicates system data to a PC using UART.



TIVA 4c123gh6pm

2. Inertial Measurement Unit (MPU6050):

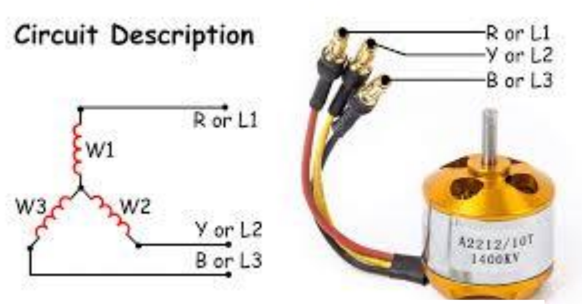
- **Description:** A 6-axis inertial measurement unit containing a 3-axis accelerometer and a 3-axis gyroscope, capable of providing motion and orientation data through an I2C interface.
- **Role:** Measures the beam's pitch angle by providing acceleration and angular velocity data, which is essential for feedback in the balancing control system.
- **Interaction:** Communicates with the TM4C123GH6PM microcontroller via the I2C protocol at an approximate sampling rate of 50–100 Hz. The microcontroller processes this data to estimate the beam angle used by the controller.



MPU 6050

3. Brushless DC Motors (A2212 1400 kV BLDC Motors):

- **Description:** High-speed brushless DC motors commonly used in drone applications, capable of producing significant thrust when paired with propellers.
- **Role:** Act as the primary actuators of the system by generating thrust at both ends of the beam to counteract angular disturbances and maintain balance.
- **Interaction:** Operated through electronic speed controllers that receive PWM signals from the microcontroller through interrupts. Motor speed variations create differential thrust to stabilize the beam.



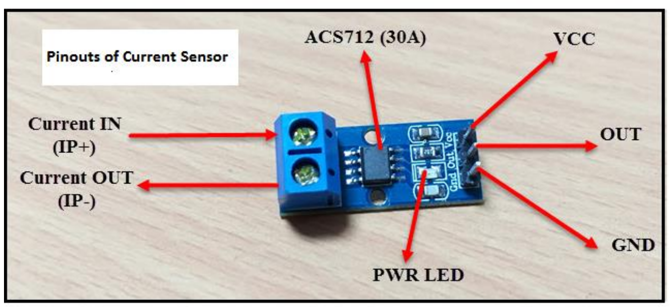
BLDC

4. Electronic Speed Controllers (30 A ESCs):

- **Description:** Power electronic devices designed to control the speed of BLDC motors by converting low-power PWM control signals into high-current motor drive signals.
- **Role:** Regulate the speed of each BLDC motor based on Interrupt signals generated by the microcontroller.
- **Interaction:** Receive Interrupt signals from GPIO pins PB6 and PB7 of the TM4C123GH6PM and supply controlled power to the BLDC motors accordingly.

5. Current Sensor (ACS712):

- **Description:** A Hall-effect based current sensor that provides an analog voltage proportional to the current flowing through the load.
- **Role:** Measures the current drawn by the BLDC motors to monitor power consumption and ensure safe system operation.
- **Interaction:** Outputs an analog voltage that is read by the ADC module of the microcontroller. The measured current values are transmitted to a PC via UART for monitoring.

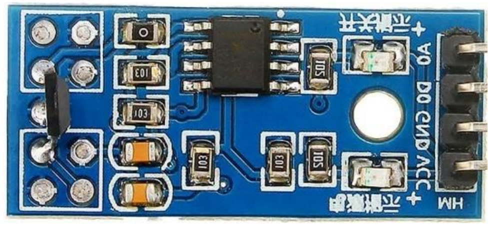


ACS712

6. Hall-Effect RPM Sensors:

- **Description:** Magnetic sensors used to detect rotational motion by generating pulses in response to a magnetic field.
- **Role:** Measure the rotational speed (RPM) of the BLDC motors, with two pulses generated per motor revolution.

- **Interaction:** Connected to GPIO pins configured for timer capture or interrupt-based counting on the microcontroller, allowing RPM calculation based on pulse frequency.



Hall Effect Sensor

7. PWM Output Pins (PB6 and PB7):

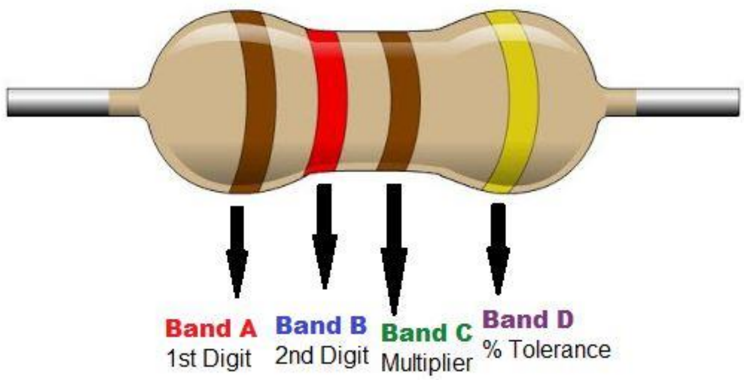
- **Description:** General-purpose I/O pins of the TM4C123GH6PM configured for hardware PWM generation using internal timer modules.
- **Role:** Generate PWM signals required by the ESCs to control motor speed and thrust.
- **Interaction:** Directly connected to the signal inputs of the ESCs, enabling real-time adjustment of motor speed by the control algorithm.

8. UART Communication Interface (UART0):

- **Description:** A serial communication interface provided by the microcontroller for data transmission and debugging.
- **Role:** Transmits real-time system data such as beam angle, motor RPM, and current to a computer for monitoring.
- **Interaction:** Communicates with a PC at a baud rate of 9600 using a serial terminal application (PUTTY). Future interaction includes data logging and plotting using Python.

9. Resistors:

- **Description:** Passive electrical components that limit current or divide voltage.
- **Role:** Used for pull-up/pull-down configurations or to limit current in input circuits.
- **Interaction:** Placed in series or parallel with buttons and sensors for reliable signal processing.



Resistor

10. Jumper Wires:

- **Description:** Flexible wires used for making temporary electrical connections.
- **Role:** Connect various electronic components on the breadboard or PCB.
- **Interaction:** Provide signal and power paths between the microcontroller, sensors, display, and motor driver.

11. PCB Design:

- **Description:** A printed circuit board fabricated for neat and reliable electrical connections.
- **Role:** Organizes and supports electronic components for robust project assembly.
- **Interaction:** Houses soldered connections for microcontroller, sensors, and motor control wiring.

12. **Wooden Structure:**

- **Description:** A rigid wooden framework used to support and mount the beam assembly, motors, and electronic components.
- **Role:** Serves as the mechanical base of the thrust-based beam balancing system, ensuring structural support and vibration reduction.
- **Interaction:** Provides stable mounting and proper alignment for the beam, BLDC motors, IMU, and control electronics to enable accurate sensing and controlled motion.

13. **Power Supply:**

- **Description:**
  A regulated DC power supply rated at 12 V output with a maximum current capacity of 60 A, suitable for high-current motor-driven systems.
- **Role:**
  Provides the primary electrical power required for operating the ESCs, BLDC motors, and associated control electronics.
- **Interaction:**
  Supplies power directly to the ESCs for motor operation while sharing a common ground with the microcontroller and sensor circuitry to ensure proper signal reference.

“PINOUTS OF THE COMPONENTS”

- This section describes the pin-level connections between the TM4C123GH6PM microcontroller and the various sensors, actuators, and communication interfaces used in the thrust-based beam balancing system.

Microcontroller Clock Configuration:

- System Clock Frequency: 16 MHz
- Clock Source: Internal oscillator

The microcontroller operates at a fixed clock frequency of 16 MHz to ensure consistent timer operation, PWM generation, and control loop execution.

1. PWM Output Connections (ESC Control):

Microcontroller Pin	Peripheral Function	Connected Device
PB6	PWM (Timer0A)	ESC for BLDC Motor 1
PB7	PWM (Timer1A)	ESC for BLDC Motor 2

These pins are configured to generate PWM signals compatible with standard RC ESCs. The PWM duty cycle controls the speed of each BLDC motor along with the interrupts and hence the thrust applied at each end of the beam.

2. IMU Interface (MPU6050 – I2C Communication):

Microcontroller Pin	**Peripheral Function**	**Connected Device**
PB2	I2C0 SCL	MPU6050 SCL
PB3	I2C0 SDA	MPU6050 SDA

The MPU6050 is interfaced with the microcontroller using the I2C protocol. This interface is used to read accelerometer and gyroscope data at a sampling rate of approximately 50–100 Hz.

3. RPM Measurement (Hall-Effect Sensors):

Microcontroller Pin	Peripheral Function	Connected Device
GPIO (Timer Capture / Interrupt)	Pulse Input	Hall Sensor – Motor 1
GPIO (Timer Capture / Interrupt)	Pulse Input	Hall Sensor – Motor 2

Each Hall-effect sensor generates two pulses per motor revolution. The microcontroller counts these pulses using timer capture or interrupt-based methods to calculate motor RPM.

4. Current Measurement (ACS712 Sensors):

Microcontroller Pin	Peripheral Function	Connected Device
ADC Channel	Analog Input	ACS712 (Motor 1 Current)
ADC Channel	Analog Input	ACS712 (Motor 2 Current)

The ACS712 sensors provide an analog voltage proportional to motor current. These signals are read using the ADC module of the microcontroller for monitoring and safety analysis.

5. Serial Communication (UART Interface):

Microcontroller Pin	Peripheral Function	Connected Device
PA0	UART0 RX	PC (USB-to-Serial)
PA1	UART0 TX	PC (USB-to-Serial)

UART0 is configured at a baud rate of 9600 bps to transmit real-time system data such as beam angle, motor RPM, and current values to a PC for monitoring using PuTTY and python's GUI.

6. Power and Ground Connections:

- All sensors and control circuitry share a common ground with the microcontroller.
- Logic-level power is supplied according to component requirements.
- ESCs are powered separately and receive only control signals from the microcontroller.

"Control Logic Of PID Controller"

The PID control logic is implemented to regulate motor behavior using real-time feedback and improve system stability. The controller operates by continuously comparing the desired setpoint with the measured system response and generating a corrective control signal for the motor.

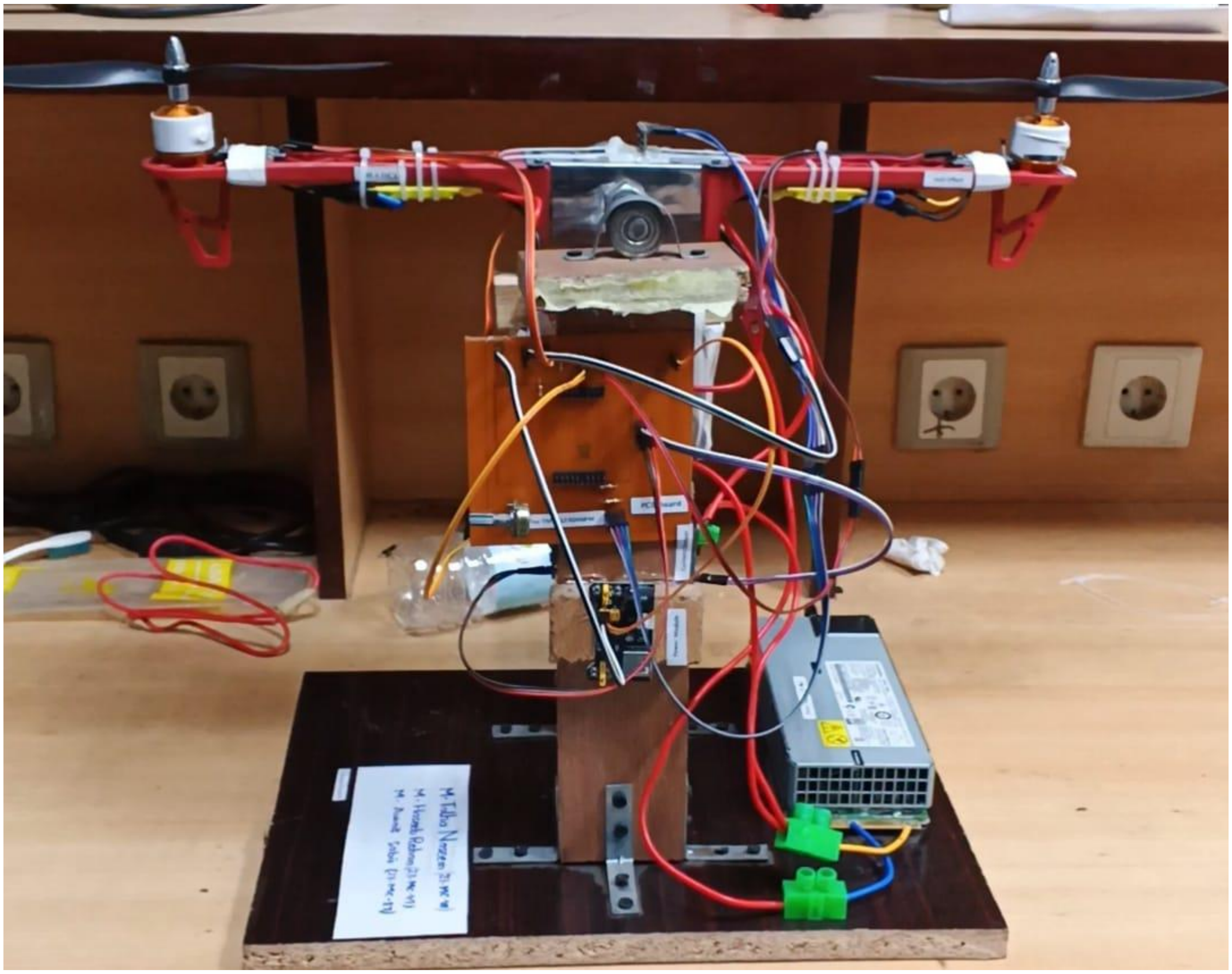
At system startup, a slow-start mechanism is applied to prevent sudden motor acceleration and mechanical stress. During an initial predefined delay period, the controller outputs a fixed low motor pulse, allowing the system to reach a safe operating condition before closed-loop control begins.

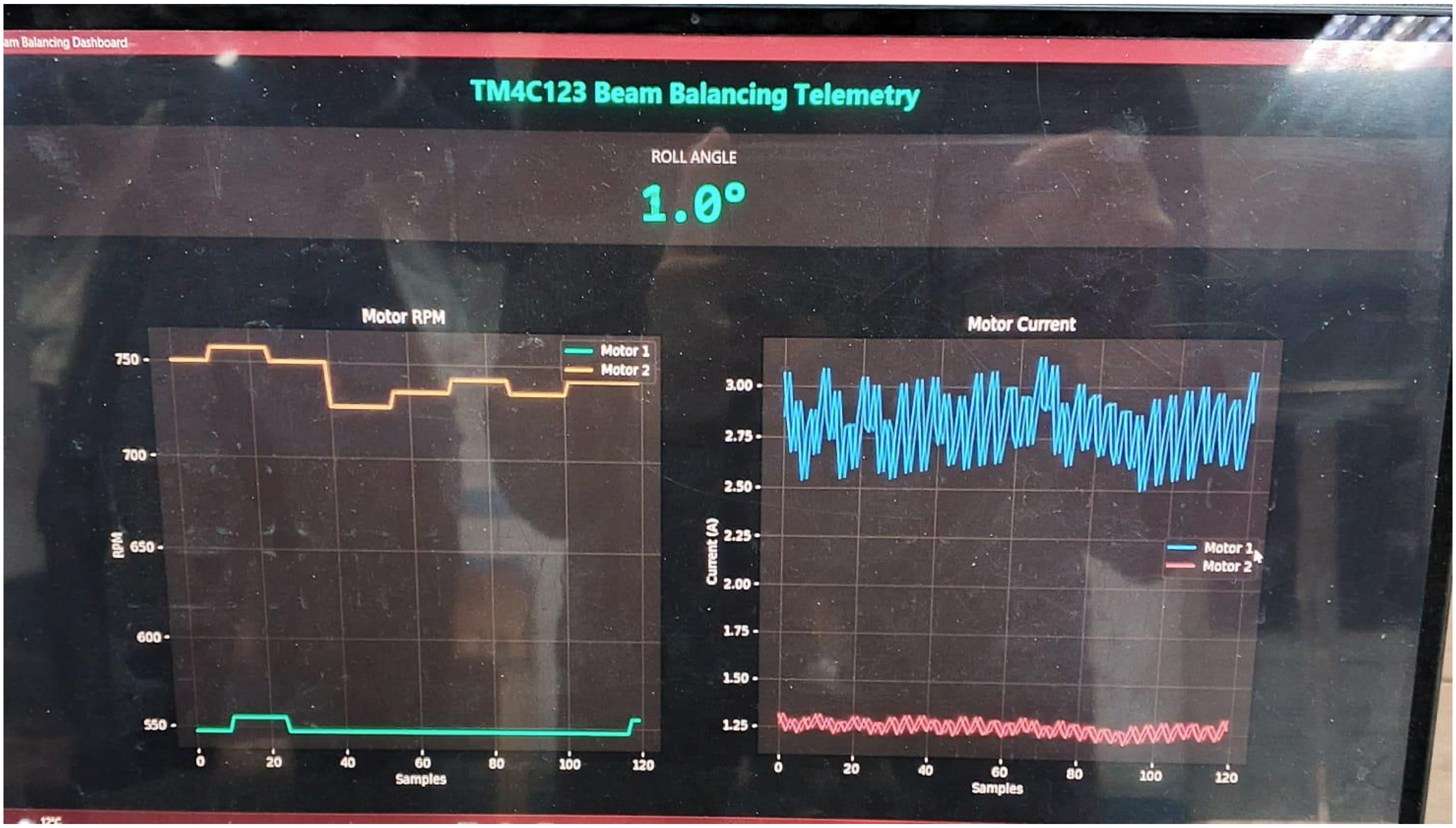
Once the initial delay has elapsed, the PID controller becomes active. The proportional term generates an output proportional to the current error between the setpoint and the measured value, providing immediate corrective action. The integral term accumulates the error over time to eliminate steady-state offset, while the derivative term predicts system behavior based on the rate of change of error, improving response smoothness and reducing oscillations.

The PID output is computed at each control cycle using the elapsed sampling time. To ensure safe operation, the controller output is saturated within predefined limits, preventing excessive control signals that could damage the motor or ESC. The final PID output is then used as the motor control command.

This structured control logic ensures stable system response, smooth startup behavior, and reliable closed-loop performance suitable for real-time embedded control applications.

“Actual Images Of The Project”





am Balancing Dashboard
TM4C123 Beam Balancing Telemetry
ROLL ANGLE
1.0°
Motor RPM
Motor 1
Motor 2
750
700
650
600
550
RPM
0
20
40
60
80
100
120
Samples
Motor Current
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
Current (A)
Motor 1
Motor 2
0
20
40
60
80
100
120
Samples

“Steps To Set Up & Use the Project”

1) **System Inspection**
   The completed system was inspected to ensure all electrical connections, sensor interfaces, and mechanical components were securely mounted and functioning correctly.
2) **Power Supply Initialization**
   A regulated 12 V DC power supply was connected to the ESC, while the microcontroller and sensors were powered through appropriate voltage regulation using power module. A common ground was maintained across all components.
3) **Firmware Execution**
   The embedded firmware was executed on the Tiva C TM4C123GH6PM microcontroller. During initialization, GPIOs, timers, interrupts, communication interfaces, and the PID controller parameters were configured.
4) **Safe Start-Up Phase**
   Upon powering the system, an initial delay-based slow start was applied. During this phase, a low motor control pulse was generated to prevent sudden motor acceleration and ensure safe operation.
5) **Sensor Feedback Acquisition**
   Hall-effect sensors continuously generated pulse signals corresponding to motor rotation. These pulses were processed using GPIO interrupts and timer-based logic to compute real-time motor RPM.
6) **PID-Controlled Operation**
   After the initial startup delay, the PID controller became active. The controller continuously compared the desired setpoint with the measured RPM and adjusted the motor control signal accordingly.
7) **Data Monitoring and Logging**
   Real-time system data, including RPM values and control output, was transmitted to a PC through UART communication. The data was observed using a serial terminal and later used Python for plotting and analysis.
8) **Performance Evaluation**
   The system response was evaluated by observing stability, transient behavior, and steady-state performance under different operating conditions.

“GPIO PINS USED”

- Total no of GPIO pins used are **9**.

- **HALL EFFECT**:

Pins of Hall Effect	Corresponding GPIO Pins
Output 1	PB2
Output 2	PB3

- **ESC**:

Pins of Esc	Corresponding GPIO Pins
Output 1	PB6
Output 2	PB7

- **MPU6050**:

Pins of MPU	Corresponding GPIO Pins
SCL	PA6
SDA	PA7

- **CURRENT SENSORS**:

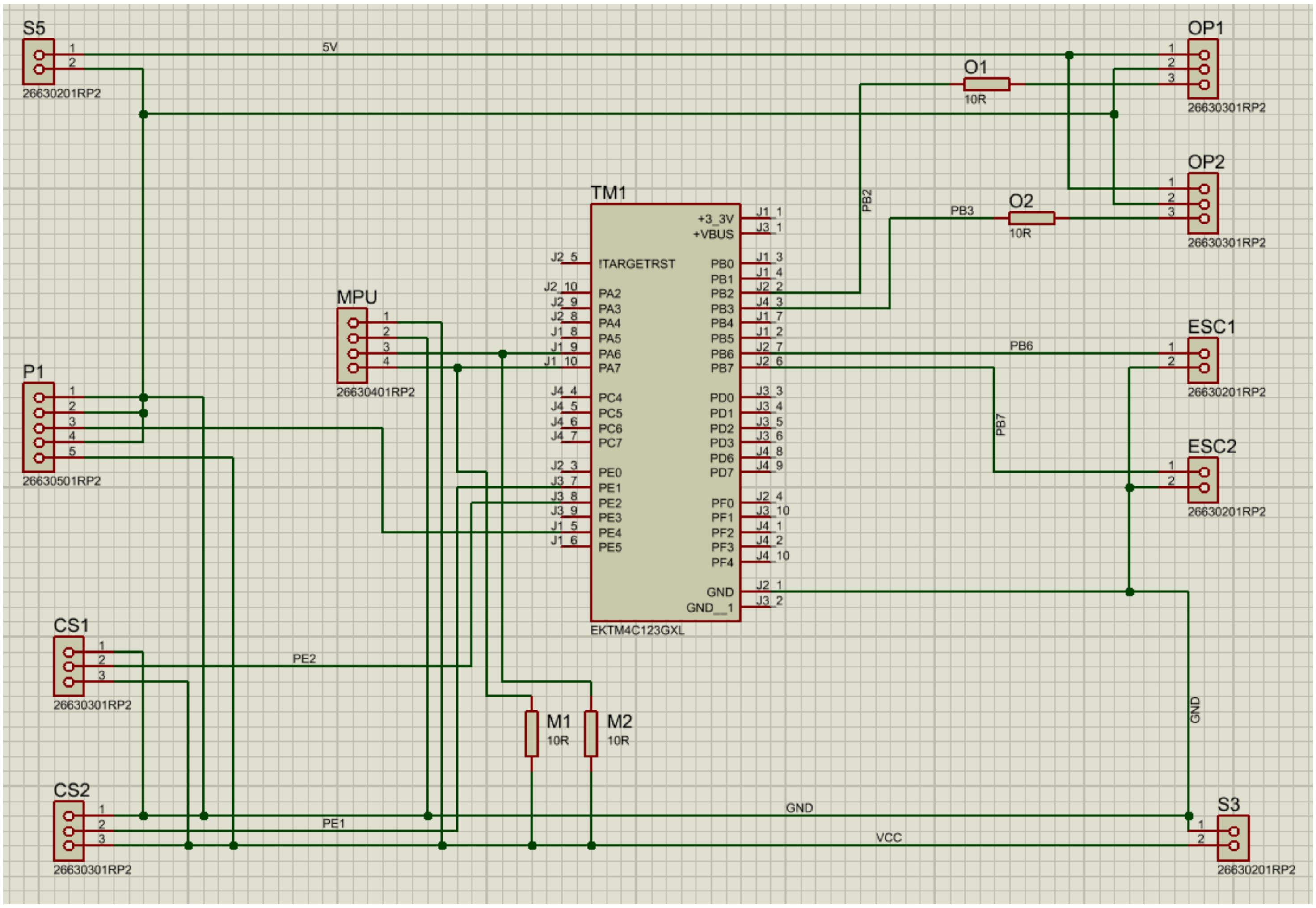
Pins of Current Sensors	Corresponding GPIO Pins
Output 1	PE2
Output 2	PE1

- **POTENTIOMETER**:

Pin of potentiometer	Corresponding GPIO Pins
Output	PE4

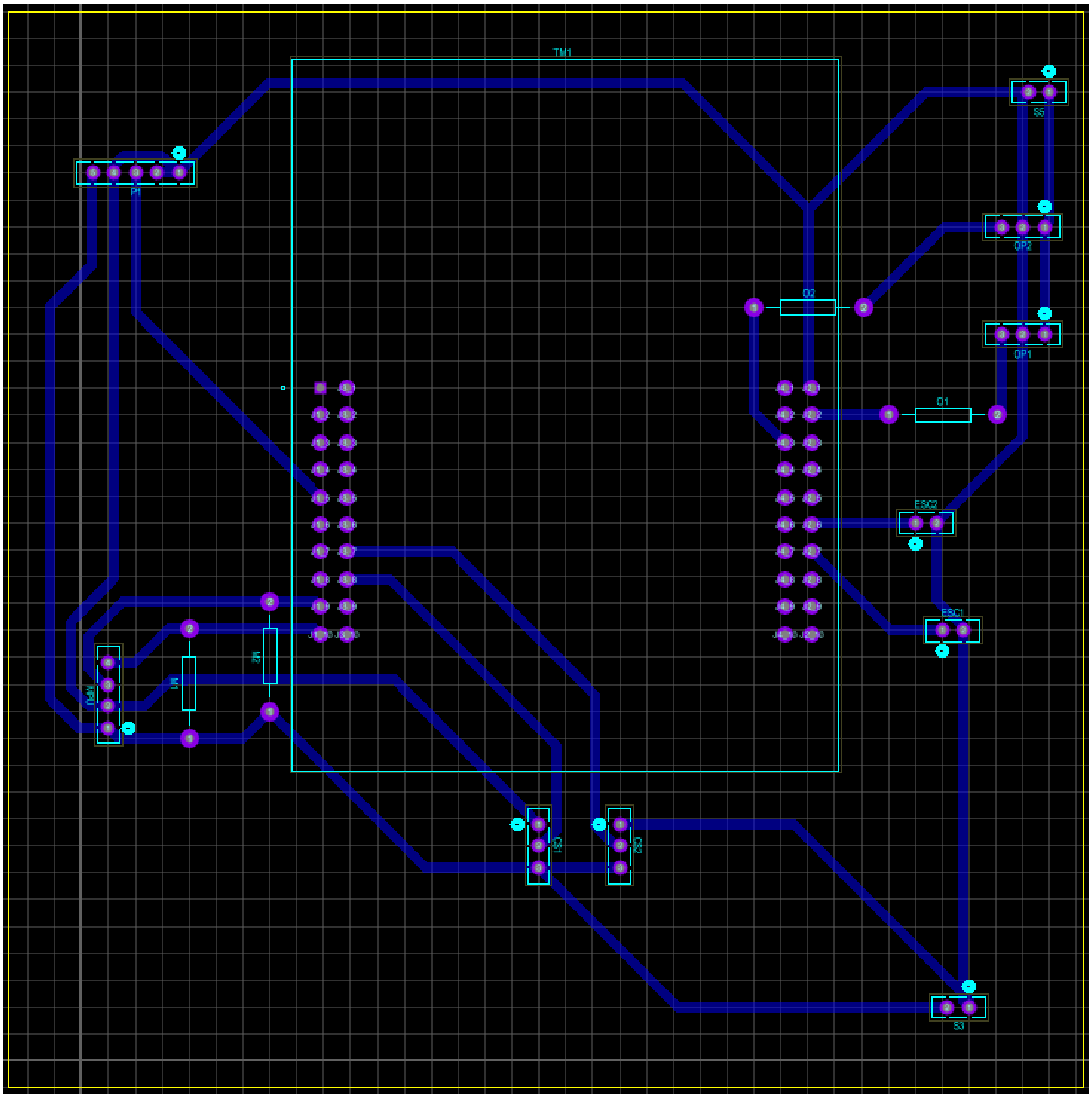
“CIRCUIT DIAGRAM AND SCHEMATIC OF PCB”

Circuit Diagram:



Circuit Diagram

Schematic of PCB:



Schematic Of PCB

“BILL OF MATERIALS”

Sr.No	COMPONENT	DESCRIPTION	MODEL	QUANTITY	PRICE	VENDOR	CONTACT	WEBSITE
1	Microcontroller (Tiva C Series)	A 32-bit ARM Cortex-M4 based microcontroller	TM4C123GH6PM	1	9000	Electronic Solution, Hall Road	0321 9469829	https://www.electronicsolution.pk/
2	Current Sensor	±20A range; 100mV/A sensitivity; 66-185mV/A scale; 1.2mΩ internal conductor resistance; 5V DC supply	ACS712	2	640	IC Shop, Hall Road	0322 4401074	https://theicshop.pk/
3	MPU	MEMS 3-axis gyro (±250 to ±2000°/s) and 3-axis accel (±2g to ±16g); 16-bit ADC per channel; I2C (400kHz)	IMU 6050	1	350	IC Shop, Hall Road	0322 4401074	https://theicshop.pk/
4	BLDC Motor	1400 KV (RPM/V); Max efficiency 80%; Max current 12A/60s; 3.17mm shaft diameter	A2212	2	2300			
5	ESC	30A continuous burst current; 5V/2A Linear ;1ms–2ms pulse width range.	30A	3	3800	Electronic Solution, Hall Road	0321 9469829	https://www.electronicsolution.pk/
6	Hall-Effect Sensor	3.8V to 24V operating range; 50ns rise/150ns fall time; bipolar or unipolar magnetic latching.	S493 & LM393	2	300	Digilog Electronics	3124002221	https://digilog.pk/
7	Adapter 1	DC power supply with 12 V output and a current of 4A.	12V, 4 Ampere	1	600			
8	Potentiometer	Rotary potentiometer and 10 kΩ linear taper.	10KΩ	2	40	Digilog Electronics	3124002221	https://digilog.pk/
9	Resistors	3.3 and 4.7 kΩ resistance, linear taper, with a 0–25 V voltage range and 0.5 W.	3.3KΩ, 4.7KΩ	2+2	10			
10	Jumper Wires	Flexible jumper wires with insulated PVC coating.	Nill	Bunch of Wires	450	IC Shop, Hall Road	0322 4401074	https://theicshop.pk/
11	PCB Design	Printed Circuit Board (PCB) with copper traces and FR4 material.	Single Sided	1	450	Hall Road	3234024338	N/A
12	Mechanical Beam	18.5" total length; Dual drone arm construction; Stainless steel pivot; Ball-bearing center axis, Propellers	DJI F550 (Kit)	2	2730	Iqbal Kharadia, Bernth Road	3229304593	N/A
13	Wood	For supporting the base and mounting IR sensors on it.	N/A	1	350	Malik Furniture	3224373718	N/A
14	Nails & Clamps	Used for assembling the wood.	N/A	10	320	Malik Furniture	3224373718	N/A
15	Wood Glue & Hot Glue	Used to assemble the woods for sorting	N/A	10	150	Malik Furniture	3224373719	N/A
16	Power Module	Out: 3.3/ 5V,Inp: 6.5-12 providing up to 700mA current.	MB102	1	220	IC Shop, Hall Road	0322 4401074	https://theicshop.pk/
17	Power Supply	12V DC / 60A (720W); ±10% voltage trim; forced-air cooled; <120mVp-p ripple.	N/A	1	2400	Epro	42237233587	https://epro.pk/
18	OptoCoupler	Encoder Speed sensor; 3.3-5 V DC	N/A	2	240	Epro	42237233587	https://epro.pk/
19	Header Pins	Male and female	N/A	8	160	Epro	42237233587	https://epro.pk/
20	Magnets	Small round magnets	N/A	4	120	Epro	42237233587	https://epro.pk/
21	Petrol	N/A			1200	Epro	42237233587	https://epro.pk/

SUBTOTAL: 25,830

- **RECIPTS:**

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Cashier: CASHIER 02/05/2025
Invoice No.: 30441 2:41PM
M/S: Cash Sales
Remarks:

Descriptions	Qty	Rate	Disc	Amnt
SG 90 SERVO MOTOR	4	250.00	0.00	1000.00
MALE TO FEMALE 10CM JUMPER WIRE	1	150.00	0.00	150.00

Total Rs. :	1150.00
Total Disc: Rs. :	0.00
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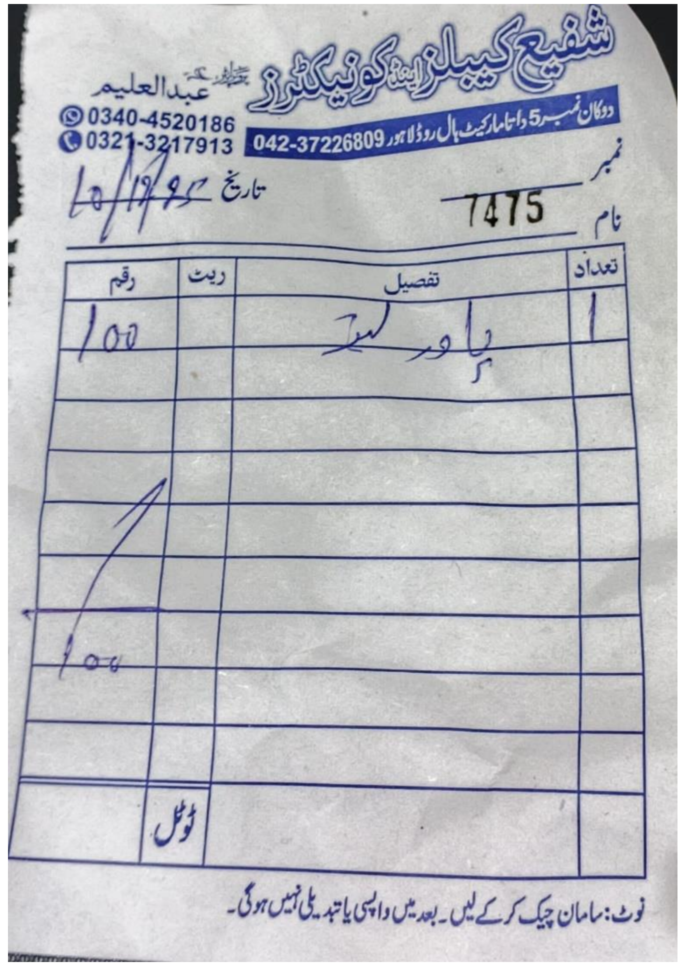

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تاریخ
نمبر 7475
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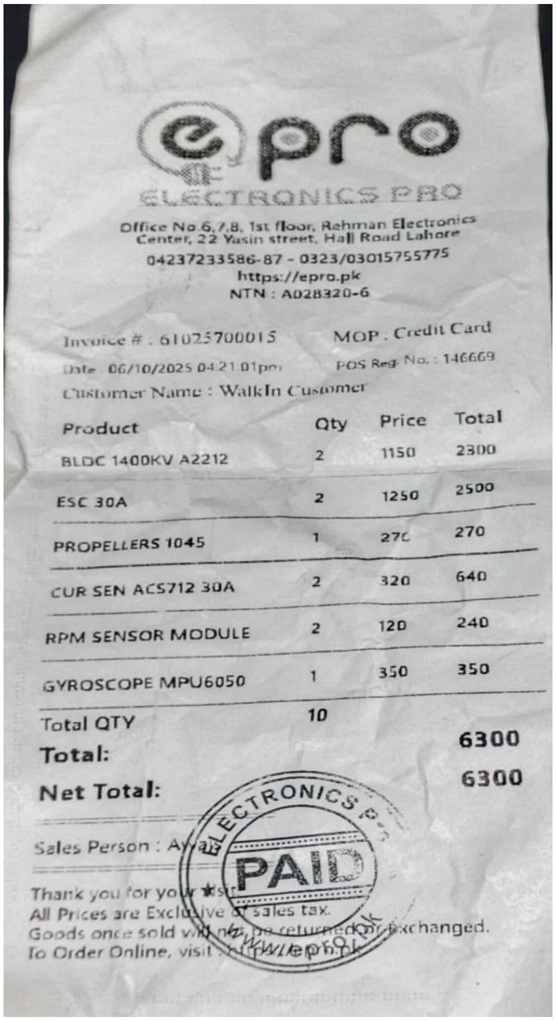

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Invoice # . 61025700015 MOP . Credit Card
Date 06/10/2025 04:21:01pm POS Reg. No. : 146669
Customer Name : WalkIn Customer

Product	Qty	Price	Total
BLDC 1400KV A2212	2	1150	2300
ESC 30A	2	1250	2500
PROPELLERS 1045	1	270	270
CUR SEN ACS712 30A	2	320	640
RPM SENSOR MODULE	2	120	240
GYROSCOPE MPU6050	1	350	350

Total QTY 10
Total: 6300
Net Total: 6300

Sales Person : Ay...

Thank you for your visit
All Prices are Exclusive of sales tax.
Goods once sold will not be returned or exchanged.
To Order Online, visit https://epro.pk

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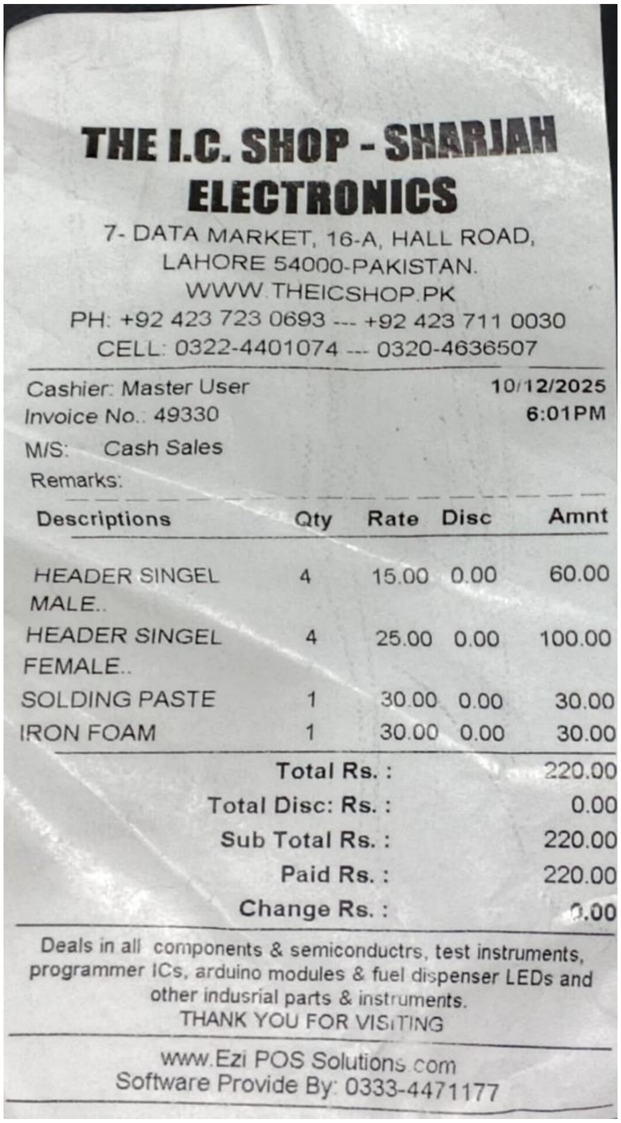
SALE

Inv#: 13548 Date: 28/Apr/2025
Type: CASH Wk#: 1
Cust Name: CASH SALES

S#	Item Name	Qty	Rate	Total
1	Female To Female 10cm	1	130.00	130

T Qty: 1.00 G Total 130
Discount: 0
Net Total 130

Goods Once Sold Will Not Be Returned Or Exchanged
Prepaired By: Mirza Soban



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Cashier: Master User 10/12/2025
Invoice No.: 49330 6:01PM
M/S: Cash Sales
Remarks:

Descriptions	Qty	Rate	Disc	Amnt
HEADER SINGEL MALE..	4	15.00	0.00	60.00
HEADER SINGEL FEMALE..	4	25.00	0.00	100.00
SOLDING PASTE	1	30.00	0.00	30.00
IRON FOAM	1	30.00	0.00	30.00

Total Rs. : 220.00
Total Disc: Rs. : 0.00
Sub Total Rs. : 220.00
Paid Rs. : 220.00
Change Rs. : 0.00

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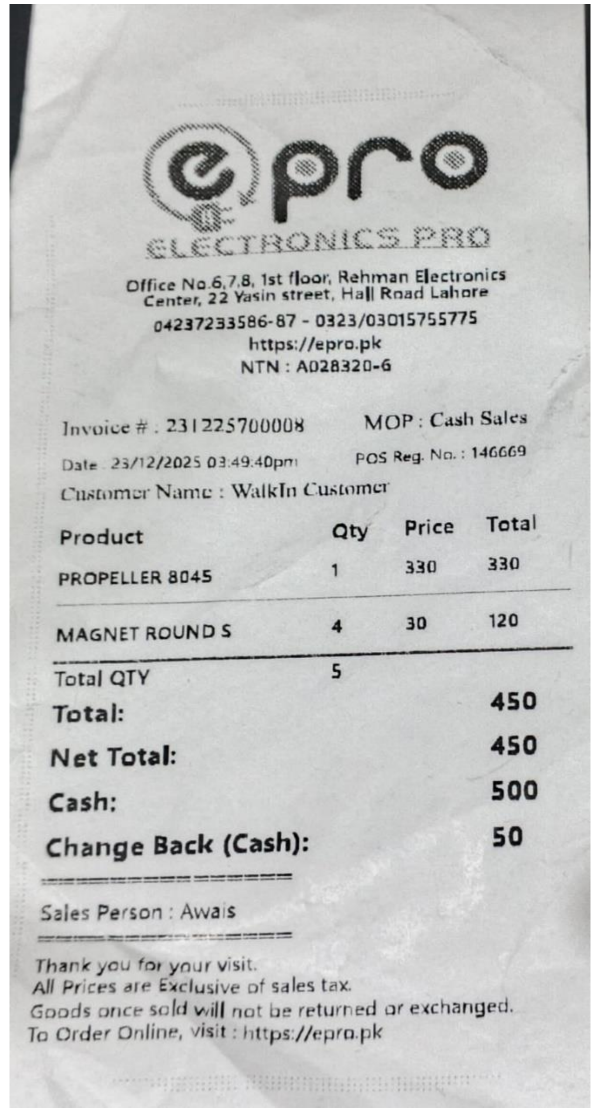

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NTN : A028320-6

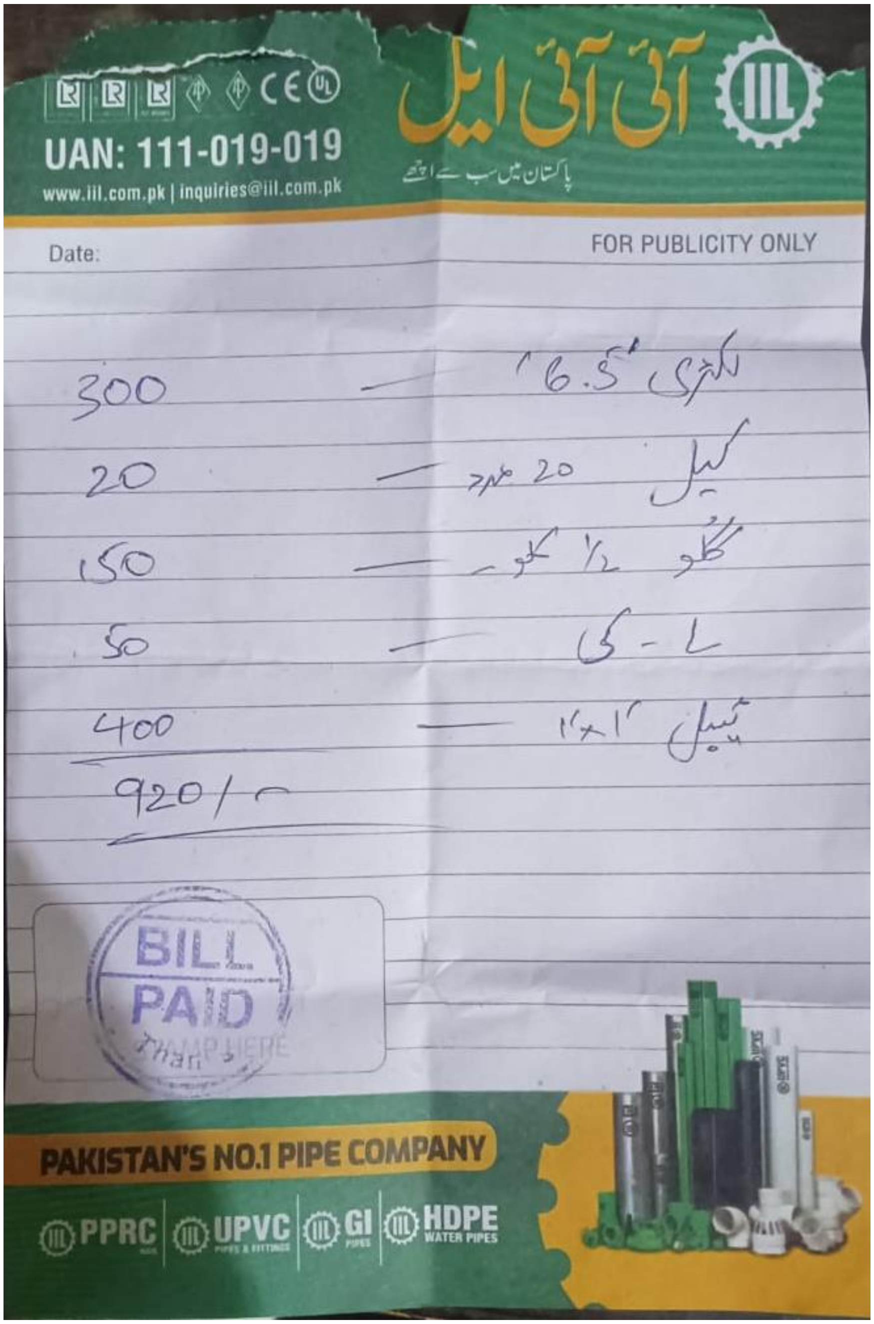
Invoice # : 231225700008 MOP : Cash Sales
Date : 23/12/2025 03:49:40pm POS Reg. No. : 146669
Customer Name : WalkIn Customer

Product	Qty	Price	Total
PROPELLER 8045	1	330	330
MAGNET ROUND S	4	30	120

Total QTY 5
Total: 450
Net Total: 450
Cash: 500
Change Back (Cash): 50

Sales Person : Awais

Thank you for your visit.
All Prices are Exclusive of sales tax.
Goods once sold will not be returned or exchanged.
To Order Online, visit : https://epro.pk



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Date:

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300 —	لکڑی '6.5
20 —	کیل 20 عدد
150 —	گلو 1/2 کلو
50 —	L - کی
400 —	ٹیبل 1'x1'
920/-	

BILL PAID

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“CONCLUSIONS”

- The thrust-based beam balancing system successfully demonstrated the practical application of embedded systems and feedback control for stabilizing an inherently unstable mechanical system.
- Using the Tiva C Series TM4C123GH6PM microcontroller operating at 16 MHz, the system performed real-time sensor acquisition, control computation, and actuator control using a bare-metal programming approach.
- The integration of the MPU6050 inertial measurement unit via the I2C interface enabled continuous estimation of the beam’s pitch angle at a sampling rate of approximately 50–100 Hz.
- Differential thrust generated by two A2212 1400 kV BLDC motors, controlled through 30 A ESCs using PWM signals, provided effective actuation for beam stabilization.
- Motor performance was monitored through Hall-effect based RPM measurement and ACS712 current sensing, allowing observation of system behavior and ensuring safer operation.
- Serial communication via UART0 enabled real-time monitoring of system parameters on a PC using a serial terminal, with provision for future graphical analysis using Python.

“REFERENCES”

1. CHATGPT
2. Google

“CREDITS”

1. **2023-MC-13**
2. **2023-MC-19**

- Roll no. 13 helped us in the debugging and making of the PCB, while roll. No 19 helped us in some circuit debugging and some PID debugging.